



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 4 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A, B
Submission Date:	18 th December, 2025		Maximum Marks:	20	
Name:			Registration Number:		

Important Instructions / Guidelines:

- Do your own work, PLAGARISM will be graded as ZERO
- No late submission.

Question 1:

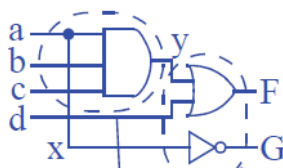
[CLO2-PLO3-C5]

[20 Marks]

Design and implement the following function using any number of 3-input 2-output CLBs using switch matrix.

$$(a,b,c,d) = abc + d$$

$$(a) = a'$$



Inputs			Outputs		Inputs			Outputs	
a	b	c	x	y	y	d	x	F	G
0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	1	0	0
0	1	0	0	0	0	1	0	1	1
0	1	1	0	0	0	1	1	1	0
1	0	0	1	0	1	0	0	1	1
1	0	1	1	0	1	0	1	1	0
1	1	0	1	0	1	1	0	1	1
1	1	1	1	1	1	1	1	1	0

